One-Shot Domain Calibration of MoE Routers via Gradient Bias Steering

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Abstract

Sparse Mixture-of-Experts (MoE) language models scale capacity by routing each token to a small subset of experts. However, a router trained on broad pretraining mixtures is not necessarily calibrated for a narrow deployment domain such as mathematics, chemistry, or history. We propose *Domain-Specific Router Adaptation* (DCRB), a one-shot post-training method that adapts a frozen MoE model by adding a small domain-specific bias to each router gate. The bias is computed from a 100-example calibration set using the gradient of the loss with respect to the router bias, accumulated only on answer tokens. After calibration, the bias is materialized as the standard bias term of the router linear layer, so inference uses the original forward path and does not require additional routing logic.

Experiments on Qwen1.5-MoE-A2.7B and Qwen3-30B-A3B show that this simple intervention can improve domain accuracy while preserving general behavior. On Qwen1.5-MoE-A2.7B, DCRB improves college mathematics by $+5.00$ percentage points, college chemistry by $+3.00$ points, and high-school U.S. history by $+2.45$ points, with no degradation on the held-out `all_part` subset for mathematics and chemistry. On Qwen3-30B-A3B, the same fixed calibration rule improves all four mathematics subsets or keeps them unchanged while preserving `all_part`. Routing diagnostics show that the bias changes expert preferences locally: top-K overlap remains high, while activation entropy decreases without collapsing to a single-expert regime. These results position router-bias calibration as a practical resource-efficient adaptation primitive for MoE LLMs.

1 Introduction

Sparse Mixture-of-Experts (MoE) architectures are a standard mechanism for scaling language models under constrained inference budgets. Instead of applying one dense feed-forward network to every token, an MoE layer routes each token to only K experts from a larger pool. This conditional computation increases total model capacity while keeping the number of activated parameters per token relatively small [17, 15, 8, 6]. Recent open-weight models such as Qwen1.5-MoE-A2.7B and Qwen3-30B-A3B make this setting practically important for domain-specific deployment [1, 19].

The router is also a potential adaptation bottleneck. During pretraining, it is optimized on broad data mixtures and learns a general-purpose allocation policy. When the model is used in a specialized domain, such as mathematics or chemistry, the router may underuse experts that are useful for that domain and overuse more generic experts. This is undesirable when the relevant knowledge is already present in the frozen model but is not consistently exposed by the default top-K routing policy. Full fine-tuning can adapt the model, but it is expensive for MoE LLMs and can disturb the pretrained interaction between routers, experts, and load balancing. Parameter-efficient fine-tuning reduces the number of trainable parameters but still requires iterative optimization and additional trainable modules [12, 10].

We study a more minimal adaptation mechanism: can a frozen MoE LLM be specialized by changing only the additive bias of its router gates? Our method, *Domain-Specific Router Adaptation* (DCRB), performs one calibration pass over a small labeled set from the target domain. For each MoE layer and expert, it estimates whether increasing that expert's router bias would locally reduce the answer loss. The resulting importance vector is normalized, scaled using the router's own logit statistics, and added to the gate bias. No pretrained weight matrix, expert parameter, attention parameter, or embedding parameter is updated.

The key implementation detail is that the calibrated vector is not used as a separate runtime lookup table. It is materialized directly as the bias term of the router linear layer: if the original router has no bias, a zero bias vector is created and the calibrated offset is added to it. Therefore, the standard router forward pass computes the modified logits automatically, and inference does not require a custom routing procedure. The number of activated experts remains unchanged.

This paper makes one focused contribution: we formulate and evaluate one-shot domain adaptation of MoE LLMs as *additive gate-bias calibration*. The method is designed for resource-efficient settings: it uses 100 calibration examples per domain, requires one forward/backward pass, stores only one scalar per expert per MoE layer, and introduces no additional expert computation at inference time. The experimental section emphasizes absolute baseline accuracies, subject-level results, calibration protocol, resource accounting, and routing behavior, rather than presenting the method as a generic fine-tuning replacement.

2 Related Work

Sparse MoE language models. Mixture-of-Experts models route inputs to a subset of expert networks and date back to adaptive mixtures of local experts [14]. Modern sparse language models scale this idea to transformer feed-forward blocks. GShard, Switch Transformer, and GLaM show that conditional computation can increase total capacity while keeping per-token computation sparse [15, 8, 6]. DeepSeekMoE and Qwen-family MoE models further demonstrate the practical value of routed expert architectures for large-scale LLMs [5, 1, 19]. Our work does not modify MoE pretraining; it targets post-training domain adaptation of already trained MoE routers.

Efficient adaptation. Full fine-tuning updates many parameters and can be impractical for large MoE LLMs. Parameter-efficient fine-tuning methods such as LoRA reduce adaptation cost by training small low-rank modules, and recent surveys cover a broad family of PEFT techniques [12, 10]. These methods are powerful but still require iterative optimization and persistent trainable modules. DCRB instead keeps all pretrained weights frozen and adds only a calibrated router-gate bias.

Expert pruning and routing modification. Several works modify expert usage after pretraining. Expert pruning and compression methods remove or merge redundant experts to improve efficiency [9, 2, 3]. Dynamic routing methods adapt the number of active experts to input difficulty, showing that harder inputs may benefit from broader expert access [13]. Closer to our setting, C3PO performs test-time expert re-mixing, Ban&Pick adjusts routing by promoting or suppressing experts, SteerMoE steers behavior through expert activation and deactivation, and online rewiring updates routing offsets during generation [16, 4, 7, 18]. These methods demonstrate that expert utilization remains adjustable after pretraining. DCRB differs by using a fixed offline calibration step and by implementing the result as a standard gate bias, avoiding dynamic control logic at inference time.

3 Method

3.1 MoE routing with an additive gate bias

Consider an MoE layer ℓ with N_ℓ routed experts. For token t, the router computes logits

$$z_{\ell,t} = h_{\ell,t} W_\ell^r + b_\ell^r, \qquad z_{\ell,t} \in \mathbb{R}^{N_\ell}, \tag{1}$$

where $h_{\ell,t}$ is the token hidden state and b_ℓ^r is the router-gate bias. If the pretrained implementation has no bias term, we treat $b_\ell^r = 0$ and create it during calibration.

The router selects

$$\mathcal{A}_{\ell,t} = \mathrm{TopK}(z_{\ell,t}, K_\ell) \tag{2}$$

and computes weights over the selected experts:

$$w_{\ell,t,i} = \frac{\exp(z_{\ell,t,i})}{\sum_{j \in \mathcal{A}_{\ell,t}} \exp(z_{\ell,t,j})}, \qquad i \in \mathcal{A}_{\ell,t}. \tag{3}$$

Let $e_{\ell,i}(h_{\ell,t})$ be the output of expert i. The MoE output is

$$o_{\ell,t} = \sum_{i \in \mathcal{A}_{\ell,t}} w_{\ell,t,i} e_{\ell,i}(h_{\ell,t}). \tag{4}$$

For a target domain d, DCRB computes an additive offset $\Delta b_\ell^{(d)}$ and applies it once:

$$b_\ell^r \leftarrow b_\ell^r + \Delta b_\ell^{(d)}. \tag{5}$$

Equivalently, the calibrated router uses

$$\widetilde{z}_{\ell,t} = h_{\ell,t} W_\ell^r + b_\ell^r + \Delta b_\ell^{(d)}. \tag{6}$$

All pretrained weight matrices are unchanged. At inference time, the model still calls the original router linear layer; the calibrated offset is simply part of its bias.

3.2 Expert importance as a gradient with respect to router bias

Let $\mathcal{D}_{\text{cal}}^{(d)}$ be a 100-example calibration set from domain d. Calibration uses teacher forcing and masks the loss so that gradients are accumulated only on answer tokens, not on prompt tokens. Let $m_t \in \{0, 1\}$ indicate whether token t is an answer token. For the next-token loss L, define

$$g_{\ell,t} = \frac{\partial L}{\partial o_{\ell,t}}. \tag{7}$$

The calibrated quantity is the additive bias $b_{\ell,i}^r$. Since $z_{\ell,t,i}$ depends additively on $b_{\ell,i}^r$,

$$\frac{\partial z_{\ell,t,i}}{\partial b_{\ell,i}^r} = 1. \tag{8}$$

Thus the relevant derivative is the derivative of the loss with respect to the router logit. For an active expert $i \in \mathcal{A}_{\ell,t}$, differentiating Eq. (4) through the softmax weights gives

$$\frac{\partial o_{\ell,t}}{\partial z_{\ell,t,i}} = w_{\ell,t,i} \left(e_{\ell,i}(h_{\ell,t}) - o_{\ell,t} \right). \tag{9}$$

The term $o_{\ell,t}$ appears because increasing one selected expert's logit changes the normalized softmax weights of all selected experts.

We define the domain importance of expert i in layer ℓ as the negative gradient direction:

$$S_{\ell,i} = - \sum_{t:\, m_t = 1} g_{\ell,t}^\top w_{\ell,t,i} \left(e_{\ell,i}(h_{\ell,t}) - o_{\ell,t} \right). \tag{10}$$

A positive $S_{\ell,i}$ means that increasing the bias of expert i is locally aligned with reducing the answer-token loss. In the main experiments, we compute and apply the bias for all routed experts in all MoE layers. Candidate scoring over larger expert sets is performed only during calibration and does not change inference sparsity.

3.3 Normalization and automatic scale

Raw scores differ in magnitude across layers. We normalize each layer independently:

$$\bar{S}_\ell = \frac{1}{N_\ell} \sum_{j=1}^{N_\ell} S_{\ell,j}, \tag{11}$$

$$\widehat{s}_{\ell,i} = \frac{S_{\ell,i} - \bar{S}_\ell}{\max_j \left| S_{\ell,j} - \bar{S}_\ell \right| + \epsilon_{\text{num}}}. \tag{12}$$

This centers the importance vector and maps it to approximately $[-1, 1]$. The final offset is

$$\Delta b_{\ell,i}^{(d)} = \varepsilon_{\ell}^{(d)} \widehat{s}_{\ell,i}. \tag{13}$$

The main scaling rule uses the router's own logit dispersion:

$$\varepsilon_{\ell}^{(d)} = \frac{4\ \mathrm{Std}_{t,i}\left(z_{\ell,t,i}\right)}{N_{\ell}}, \qquad (t,i) \sim \mathcal{D}_{\mathrm{cal}}^{(d)}. \tag{14}$$

The factor $\mathrm{Std}(z_{\ell})$ adapts the offset to the natural logit scale of layer ℓ. The division by N_{ℓ} keeps the maximum shift small relative to the full router-logit range. The intended behavior is to adjust borderline routing decisions and mixture weights, not to overwrite the pretrained routing policy.

4 Experiments

Main question. Can a frozen MoE LLM be specialized to a target domain by one calibration pass that modifies only the additive router-gate bias?

4.1 Experimental setup

Models. We evaluate Qwen1.5-MoE-A2.7B and Qwen3-30B-A3B. Qwen1.5-MoE-A2.7B has 14.3B total parameters and 2.7B activated parameters; in our experiments it uses 24 transformer layers, 60 routed experts per MoE layer, and top-4 routing. Qwen3-30B-A3B has 30.5B total parameters, 3.3B activated parameters, 48 layers, 128 routed experts, and top-8 routing. All pretrained model weights are frozen.

Calibration data. For each domain, we use a separate 100-example calibration set from that domain. The calibration examples are disjoint from the evaluation examples. When the calibration subject and an evaluation subject have the same name, e.g. `college_mathematics`, the calibration and evaluation splits are separated by example IDs. Calibration uses the correct answer under teacher forcing, and the loss mask includes only answer tokens. Prompt tokens are used as context but do not contribute to the accumulated expert importance.

Evaluation data. We evaluate on MMLU multiple-choice subsets [11]. For mathematics, the calibration subject is `college_mathematics`, and evaluation uses `college_mathematics`, `abstract_algebra`, `elementary_mathematics`, and `high_school_mathematics`. For chemistry, calibration uses `college_chemistry`, and evaluation uses `college_chemistry` and `high_school_chemistry`. For history, calibration uses `high_school_world_history`, and evaluation uses `high_school_world_history`, `high_school_european_history`, and `high_school_us_history`. We also report `all_part`, a held-out 300-example general subset used to check whether domain calibration damages non-target behavior.

Scoring and decoding. Each MMLU question is evaluated by answer accuracy. The model receives the question and answer choices, and the predicted label is the answer option with the highest likelihood under the evaluation prompt. All reported numbers are percentages. BASE denotes the unmodified model. DCRB denotes the fixed configuration from Section 3: corrected bias-gradient importance, max-normalized scores, and the $4\,\mathrm{Std}(z)/N$ scale.

4.2 Main subject-level results

Table 1 shows mathematics results. On Qwen1.5-MoE-A2.7B, DCRB improves all four mathematics subsets. The largest improvement is on `college_mathematics`, where accuracy increases from 35.00 to 40.00. The held-out `all_part` subset is unchanged. On Qwen3-30B-A3B, the base model is substantially stronger, but the same calibration rule still improves or preserves every mathematics subset and leaves `all_part` unchanged.

Table 2 reports Qwen1.5-MoE-A2.7B on chemistry and history. The chemistry bias, calibrated on `college_chemistry`, improves both chemistry subsets and preserves `all_part`. The history bias, calibrated on `high_school_world_history`, improves two of three history subsets and also improves `all_part` by $+0.33$ points. These results show that the same one-shot router-bias procedure transfers beyond the exact calibration subject within the same domain.

Table 1: Mathematics results. The bias is calibrated on 100 non-test examples from `college_mathematics`. All values are accuracies in percent.

Model	Evaluation subset	Base	DCRB	Δ
Qwen1.5-MoE-A2.7B	college_mathematics	35.00	40.00	$+5.00$
	abstract_algebra	38.00	39.00	$+1.00$
	elementary_mathematics	40.21	41.53	$+1.32$
	high_school_mathematics	30.37	31.11	$+0.74$
	`all_part`	55.67	55.67	$+0.00$
Qwen3-30B-A3B	college_mathematics	73.00	74.00	$+1.00$
	abstract_algebra	73.00	74.00	$+1.00$
	elementary_mathematics	85.19	85.19	$+0.00$
	high_school_mathematics	65.93	66.30	$+0.37$
	`all_part`	85.00	85.00	$+0.00$

Table 2: Qwen1.5-MoE-A2.7B results on chemistry and history. Chemistry is calibrated on 100 non-test examples from `college_chemistry`; history is calibrated on 100 non-test examples from `high_school_world_history`.

Domain	Evaluation subset	Base	DCRB	Δ
Chemistry	college_chemistry	45.00	48.00	$+3.00$
	high_school_chemistry	44.83	45.81	$+0.98$
	`all_part`	55.67	55.67	$+0.00$
History	high_school_world_history	70.89	72.15	$+1.26$
	high_school_european_history	68.48	67.88	-0.60
	high_school_us_history	69.12	71.57	$+2.45$
	`all_part`	55.67	56.00	$+0.33$

4.3 Resource accounting

Calibration requires gradients, so DCRB is not a black-box inference-only method. However, it is a one-pass post-training procedure rather than iterative optimization. In our implementation, calibration is performed on one NVIDIA A800 GPU with 80GB VRAM and takes less than one minute for a 100-example calibration set. No optimizer state, learning-rate schedule, or repeated epoch over the calibration data is used.

The persistent adaptation state is extremely small. For Qwen1.5-MoE-A2.7B, the bias contains $24 \times 60 = 1440$ scalars, or 2.8KB in FP16. For Qwen3-30B-A3B, the bias contains $48 \times 128 = 6144$ scalars, or 12.0KB in FP16. At inference time, the calibrated vector is part of the router linear bias. The number of activated experts and the expert computation are unchanged.

4.4 Uncertainty of small MMLU subsets

The subject-level MMLU subsets are small, so individual changes below one percentage point should not be over-interpreted. For a subject with n examples and accuracy $\hat{p}$, the approximate binomial standard error is

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}. \tag{15}$$

For $n = 100$, the largest 95% confidence half-width is about 9.8 percentage points; for $n = 300$, it is about 5.7 points. These intervals describe uncertainty of absolute accuracies, not a paired test of the deltas. A stronger significance analysis would require paired per-example predictions and a bootstrap or McNemar test. Accordingly, we focus on the repeated positive pattern across related subsets and on the preservation of `all_part`, rather than claiming that every small cell-level delta is statistically significant.

Table 3: Persistent adaptation state per domain.

Model	Routed layers $\times$ experts	Scalars	FP16 size
Qwen1.5-MoE-A2.7B	24×60	1,440	2.8KB
Qwen3-30B-A3B	48×128	6,144	12.0KB

4.5 Router behavior

We analyze whether the calibrated bias causes local rerouting or globally overwrites the router. For each layer and token, let $\mathcal{A}_{\ell,t}$ and $\widetilde{\mathcal{A}}_{\ell,t}$ be the top-K expert sets before and after calibration. We measure the top-K overlap

$$\text{Overlap} = \mathbb{E}_{\ell,t}\left[\frac{|\mathcal{A}_{\ell,t} \cap \widetilde{\mathcal{A}}_{\ell,t}|}{K_\ell}\right], \tag{16}$$

and the entropy of expert activation frequencies within each layer. On `abstract_algebra`, differential activation maps show that the bias promotes and suppresses experts in selected layers rather than changing all layers uniformly. For Qwen1.5-MoE-A2.7B, exact top-K matches remain around 40% in the layers most affected by the bias, while partial overlap remains high. For Qwen3-30B-A3B, exact matches are higher in most layers, indicating a more conservative rerouting effect. Activation entropy decreases after calibration, meaning the model concentrates more probability mass on a narrower domain-relevant expert subset, but the entropy never collapses to zero. Thus, DCRB changes router preferences without producing a degenerate single-expert regime.

4.6 Scale sanity check

We tested several automatic scales for ε_ℓ, including IQR-based scaling, margin-based scaling, and squared-score scaling. The main $4\,\mathrm{Std}(z)/N$ scale was selected because it is conservative and stable. In the diploma-scale experiments, this scale improved Qwen1.5-MoE across mathematics, chemistry, and history while preserving MMLU-style general accuracy. Margin-based scaling can produce larger individual gains, but it is more sensitive to model architecture and can affect general accuracy more strongly. For the short workshop paper, we therefore use the conservative scale as the default and treat more aggressive variants as ablations rather than as the main method.

5 Discussion

What the results show. The results support the central hypothesis that some domain adaptation of MoE LLMs can be achieved by calibrating the router rather than updating expert weights. On Qwen1.5-MoE-A2.7B, a 100-example calibration set is sufficient to improve multiple mathematics, chemistry, and history subsets. On Qwen3-30B-A3B, where the baseline is already much stronger, the same intervention produces smaller but still positive mathematics changes and preserves `all_part`. This behavior is consistent with the intended role of DCRB: it is a lightweight calibration mechanism that is most useful when the pretrained router leaves domain-specific headroom.

Why additive gate bias is practical. The calibrated offset is inserted into the existing router-gate bias. This is different from maintaining a separate lookup table that must be queried by custom inference code. Once the offset is added, standard PyTorch linear layers already include the bias term in the forward pass. The model architecture, number of active experts, and expert weights remain unchanged. As a result, multiple domain-specialized copies can be represented by very small bias deltas rather than by separate fine-tuned checkpoints.

Limitations. DCRB requires gradient access during calibration, so it is not applicable to black-box API-only models. It also cannot add new knowledge to the model; it can only improve how the frozen model uses its existing experts. The current evaluation uses MMLU-style multiple-choice tasks and a small number of MoE architectures. Future work should evaluate generation tasks, larger calibration pools, multi-domain bias composition, and paired significance tests over per-example predictions.

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